

Dr Khalid A. Fakhro

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Professional Experience

Current Posts and Academic Affiliations:

Professional Roles

Chief Research Officer, Sidra Medicine, Doha, Qatar	Aug 2019 – present
Management Committee (CEO), Sidra Medicine, Doha, Qatar	Jan 2022 – Dec 2022
Chair of the Sidra Precision Medicine Program	Jan 2018 – present
Principal Investigator – Laboratory of Human Genetics and Genomic Medicine	Jun 2014 – present

National Leadership Roles

Board Member, Qatar Precision Health Institute	Jul 2019 – present
Board Vice Chair, Diabetes Qatar	Jul 2023 – present
Graduate Program Committee, Hamad Bin Khalifa University	Sept 2017 – present
Program Advisory Committee, Executive MHA, Univ. of Doha Science & Tech	May 2024 – present
Steering Committee, Qatar–Harvard Stem Cell Program	Jan 2022 – July 2023
Program Steering Committee, World Innovation Summit for Health	Apr 2022 – Oct 2022

Academic Appointments

Professor of Genomics & Precision Med., College of Health & Life Sci., HBKU, Qatar	Sep 2021 – present
Assoc. Professor of Genetic Med., Weill Cornell Medical College, New York and Qatar	Aug 2018 – present
Assoc. Professor of Neuroscience, WCMC, NY (Tri-Institutional Graduate school)	May 2020 – present

Current Role:

Chief Research Officer and Chair of Precision Medicine Program, Sidra Medicine Aug 2019 – present

Leading the Research and Innovation at Sidra Medicine, responsible for the development and implementation of Sidra's Research Strategy, focused on precision diagnostics, personalized therapies, life sciences innovation and rigorous implementation science. Transformed Sidra into a globally reputable, high-performance academic medical center: >500 research-active staff, >290 studies, >55 international collaborations, ~100 papers per year (mean IF: 8.7) while covering >40% of cost-basis from grants or commercial revenue (up from 5% in 2019).

Key achievements:

Developed systems-level approach to Research strategy, administration and operations, unifying governance under a new Research Steering Committee • Created a project management office (PMO) and introduced bottom-up budgeting of all research activities • Established the first competitive intramural Internal Research Fund focused on translational and implementation research • Established the human research protection program (HRPP), and successfully passed JCI HRP twice with 0 citations • Inaugurated Sidra's clinical trials office (CTO), attracting biotech and industry trials • Established Qatar's first clinical-grade WGS service, a rapid NICU-seq protocol, and a zebrafish core for functional disease modeling • Established Qatar's first GMP facility for cell and gene therapy • Established Qatar's first national cord blood bank • Inaugurated Qatar's first hospital-based Innovation Lab to accelerate commercialization & tech transfer and the first Healthcare Accelerator (with Cure by Deerfield) • Built and Directed Sidra's Human Genetics Dept focused on gene discovery & population genomics • Launched Qatar's first "Mendelian Disease Program" and Qatar's largest Autism Study "BARAKA-Qatar" • Established visiting scientist and trainee tracks, hosting >60 MS/PhD and >300 college and high school students at Sidra labs to date • Started weekly science seminars (since 2017), 3-4 high profile conferences & workshops per year, including Sidra's flagship Precision Medicine & the Future of Genomics (PMFG) meeting.

Other leadership roles held at Sidra Medicine

	Dates
Chair – Sidra’s flagship Precision Medicine & Future of Genomics Conference	Jan 2021 – Dec 2022
Chair – Sidra Internal Research Council (now, Research Steering Committee)	Oct 2020 – present
Head – Department of Human Genetics	Jul 2016 – Aug 2020
Principal Investigator – Human Genetics	Jun 2014 – Jul 2016
Member – Sidra Capital Review Committee	Jun 2019 – present
Member – Sidra Tender Committee	Jan 2016 – Dec 2019
Member – Sidra Merit Committee	Jan 2017 – Jul 2018
Member – Sidra Institutional Research Board (IRB)	Jan 2016 – Dec 2018
Member – Sidra National Development Committee	Sep 2015 – Dec 2018
Additional relevant leadership responsibilities:	

Member, Sidra Medicine Management Committee

Jan 2022–Dec 2022

Appointed by Sidra Medicine Chairperson to serve on the hospital’s Management Committee during the transition phase to appoint a new CEO. The MC’s mandate is to act as the CEO for Sidra Medicine, in charge of creating a 5-year strategy, aligning resources and staff towards the long-term sustainability goals, and ensuring Sidra Medicine achieves its vision of becoming a world class academic medical center for women and children

Board Member, Qatar Precision Health Institute (QPHI)

2020-present

Appointed to the Board by the Chairperson of Qatar Foundation. Worked with the QPMI ISAC to establish a national strategy for Precision Medicine, covering key pillars such as sequencing, bioinformatics, clinical implementation of Precision Medicine, education and mentorship, and Islamic bio-ethics.

Board Vice Chair, Diabetes Qatar

2023-present

Appointed vice chair of the Board by the Chairperson of Qatar Foundation. Leading national stakeholders and industry partners on a unified national strategy for the diagnosis, treatment and prevention of diabetes in Qatar.

Policy Steering Committee, World Innovation Summit for Health (WISH)

2021-2024

Appointed to the 2022 WISH Policy Steering Committee, chaired by the WISH CEO, to guide, influence and support the main program of the 6th global WISH summit. Selected as an Advisory Group Member for two WISH Forums: Vaccine Innovation, and Islamic Ethics and Healthcare of Children in the Age of Genomics.

Consultant, Qatar Genome Programme (QGP)

2014-2016

Appointed by Board Vice Chairperson to develop a National Genome Program framework for Qatar. Led the scoping of a short- and long-term strategy for genomics and its implementation into healthcare, liaised with local stakeholders and global partners, hired the first wave of employees to run the QGP, created a QGP publication strategy, established capacity building plans, and worked with local Universities (QU/HBKU) to start programs in genomics, precision medicine and genetic counseling to build future sustainability for this field in Qatar.

Associate Professor, Department of Genetic Medicine, Weill-Cornell Medical College

2019-present

Moved to WCM-Q from WCM-NY in 2011 to establish applied research capacity in human genomics in Qatar, within the nascent Department of Genetic Medicine. Over the last decade, was appointed to Assistant Professor (2013) and subsequently Associate Professor (2019), published landmark studies on Qatari population/disease genetics, taught medical genetics courses, mentored education city students, and executed key research agreements facilitating research between WCM-Q and local stakeholders in biomedical research.

Professor, College of Health & Life Sciences, Hamad Bin Khalifa University

2016-present

Worked with CHLS since its inception (2016) to establish graduate programs to build capacity in Precision Health in Qatar. Designed syllabus for 2 inaugural PhD programs: Genomics and Precision Medicine (GPM) and Biological and Biomedical Sciences (BBS). Supervised Masters/PhD students and taught courses on molecular biology, human genetics, population genomics, etc... Appointed by Dean for 4 consecutive terms (8 years) as the only external member of the Graduate Program Committee (GPC).

Academic Experience

Education

University of Pennsylvania, Wharton General Management Program (GMP)

Harvard University, LEAD Healthcare Leadership Immersion Program

MIT-HMS, Executive Education - Healthcare Innovation and Entrepreneurship
(co-hosted by MIT-Sloan School of Management and Harvard Medical School)

Yale University, MPhil and PhD in Human Genetics & Genomics *Thesis Advisor: Dr Richard P. Lifton*
Selected by competitive application to the *HHMI Translational Research Scholars Program*

University of Chicago, BA with Honors in Biological Sciences *Thesis Advisor: Dr Xiaoxi Zhuang*
Specializing in Cellular Biology & Molecular Genetics

Research and Teaching Experience

Research Experience: >20 years of biomedical research experience, including computational approaches for the analysis of human genetics and genome structure, and wet lab techniques and animal models (zebrafish, xenopus and mouse). Trained at internationally competitive laboratories under highly rigorous mentorship.

Teaching Experience: Passion for teaching early in life as a tutor in high school and throughout College. As a graduate student at Yale University, taught several classes to undergraduate and medical students. Through faculty appointments at both Weill Cornell Medical College and Hamad bin Khalifa University, designed the syllabus for the flagship Genomics and Precision Medicine Program from 2018, and taught classes covering a wide array of topics: genomic technologies and NGS, gene discovery in rare disease, population genomics and personalized medicine. Prepared all course material, led discussions, and wrote/graded assignments/exams. Some WCMQ lectures were recorded as well for use in New York.

Mentorship of graduate students and postdoctoral fellows

2025 – Present, Mr. Berk Mankaliye, PhD Candidate, *Weill-Cornell Medical College*, New York, NY
2025 – Present, Ms. Saba Elhag, PhD Candidate, *Hamad Bin Khalifa University*, Doha, Qatar
2025 – Present, Mr. Mohamed Abuhaliqa, PhD Candidate, *Hamad Bin Khalifa University*, Doha, Qatar
2024 – Present, Ms. Dua’a Ahmed, PhD Candidate, *Hamad Bin Khalifa University*, Doha, Qatar
2024 – Present, Ms. Amira Alwa, MSc Candidate, *Hamad Bin Khalifa University*, Doha, Qatar
2023 – Present, Mr. Shahzad Ahmed, Postdoctoral Fellow, *Sidra Medicine*, Doha, Qatar
2023 – Present, Ms. Amal Ibrahim, PhD Candidate*, *Hamad Bin Khalifa University*, Doha, Qatar
2023 – Present, Ms. Rulan Shaath, PhD Candidate*, *Hamad Bin Khalifa University*, Doha, Qatar
2022 – 2025, Ms. Ayat Khadi, PhD Candidate*, *Hamad Bin Khalifa University*, Doha, Qatar
2021 – Present, Dr. Shoaib Nawaz, Postdoctoral Fellow, *Sidra Medicine*, Doha, Qatar
2021 – 2024, Ms. Alya Al-Kurbi, PhD Candidate*, *Hamad Bin Khalifa University*, Doha, Qatar
2019 – 2025, Ms. Njoud Al-Neama, PhD Candidate, *Weill-Cornell Medical College*, New York NY
2018 – 2025, Mr. Elbay Aliyev, Postdoctoral Fellow, *Sidra Medicine*, Doha, Qatar
2018 – Present, Mrs. Omayma Al-Saie, PhD Candidate*, *University College London*, London, UK
2017 – Present, Dr. Waleed Aamer, Senior Postdoctoral Fellow, *Sidra Medicine*, Doha, Qatar
2016 – 2023, Dr. Aljazi Al-Maraghi, Senior Postdoctoral Fellow, *Sidra Medicine*, Doha, Qatar
2021 – 2023, Dr. Ilaria Poggiolini, Postdoctoral Fellow, *Sidra Medicine*, Doha, Qatar
2019 – 2022, Mr. Mohammed Kohailan, PhD Candidate*, *Hamad Bin Khalifa University*, Doha, Qatar

2018 – 2022, Mrs. Geethanjali Davos, PhD Candidate*, *Hamad Bin Khalifa University*, Doha, Qatar
 2020 – 2023, Ms. Mona Abdi, PhD Candidate*, *Hamad Bin Khalifa University*, Doha, Qatar
 2020 – 2021, Ms. Alya Al-Kurbi, Masters Candidate*, *Hamad Bin Khalifa University*, Doha, Qatar
 2018 – 2021, Ms. Amal Abdelmageed, PhD Candidate, *Hamad Bin Khalifa University*, Doha, Qatar
 2019 – 2020, Ms. Sana Al-Saafin, MA Candidate*, *Hamad Bin Khalifa University*, Doha, Qatar
 2019 – 2020, Ms. Hanifa Soueid, MA Candidate, *Hamad Bin Khalifa University*, Doha, Qatar
 2018 – 2019, Ms. Lara Gamgoum, Masters Candidate, *Hamad Bin Khalifa University*, Doha, Qatar
 2017 – 2018, Mrs. Ghada Mubarak, Masters Candidate, *Hamad Bin Khalifa University*, Doha, Qatar
 2016 – 2019, Mrs. Sahar Da'as, PhD Candidate*⁺, *Hamad Bin Khalifa University*, Doha, Qatar
 *Primary Faculty Advisor | ⁺First PhD student to graduate from College of Health & Life Sciences, HBKU

Other Relevant Programs/Appointments:

Harvard LEAD Leadership Program, *Selected by competitive application*, 2018 (Feb-Nov)

Selected to be part of a 10-month bespoke intensive Healthcare leadership program for senior leaders, developed by Harvard Business School, delivered by Harvard Business Publishing.

MIT-HMS Healthcare Innovation and Entrepreneurship, *Selected by competitive application*, 2018

Applied for and was selected as part of the inaugural class of the intensive MIT Sloane summer program on Healthcare Innovation and Entrepreneurship, designed in conjunction with Harvard Medical School.

Sidra Scientific Merit Committee, *Assigned by CRO*, 2016-2019

One of 8 board members (4 clinical and 4 research) assigned by the Chief Research Officer to review and allocate nearly \$5m per year to competitive research projects within Sidra Medicine.

Sidra's National Development Program, *Assigned by CRO/CLO*, 2016-2018

Led the creation of Sidra Research's National Development plan, a 2-year framework to transition Qatari Graduate Associates into research labs within Sidra. Interface with international universities to find opportunities for graduates to obtain higher education through competitive off-site programs.

Academic Peer-Review and Editorial Responsibilities, 2015-Present

Peer-reviewer and member of editorial board for several prominent academic journals and grant agencies on topics of genomics, rare diseases, translational medicine and bioinformatics for gene discovery.

Yale HHMI Medical Research Scholars Program, *Scholar*, 2006-2008

A 2-year highly selective Howard Hughes Medical Institute (HHMI) program designed to orient graduate students in basic science towards pressing clinical problems through intensive physiology modules, medical-research related seminars and courses, and direct physician-scientist mentorship in the clinic.

Bahrain's Crown Prince International Scholarship Program, *Recipient*, 2001-2011

The most prestigious national scholarship, given to 2 highest-ranking high school students in the country, recognizing academic excellence and leadership potential; full scholarship/stipend with no conditions.

Sigma Xi: The Scientific Research Society, *Elected Member*, 2005-Present

Nominated by University of Chicago faculty and elected to Sigma Xi (www.sigmaxi.org), the honor society for scientists and engineers that recognizes excellent scientific achievement in research.

Tomouh, *Elected Member*, 2012; *Ambassador*, 2018-Present; *Admissions Committee*, 2017-8, 2024-present

Nominated for and elected to Tomouh (www.tomouh.com), a highly selective society of achievers, which serves as a network of leaders in academic/professional fields in the Gulf Arab world, enabling achievers to excel and succeed in a variety of fields through networking, mentorship, and access to rare opportunities.

Competitive Awards and Extramural Funding

Total Research Funding to date: \$15.3m with \$9.4m as Lead PI:

1. Path to Precision Medicine Cycle 8 (**PPM8- PPM 08-0210-250037**): ANGEL AI-Driven Newborn Rapid Genomic Diagnosis for Early Lifesaving in NICU and PICU Babies. PI; Amount: \$644k
2. Breakthrough T1D (**2-SRA-2022-1258-M-B**): Development and Validation of MENA Type 1 Diabetes Genetic Risk Score PI; Amount \$300k
3. Path to Precision Medicine Cycle 7 (**PPM 07-0321-240010**) MeLAD: Machine Learning Model for Misdiagnosed and Asymptomatic Type 1 Diabetes Patients. PI; Amount: \$646k
4. Path to Precision Medicine Cycle 6 (**PPM 06-0507-230014**): Improving Diagnostic Yield in Genome-Negative Families w/ Multi-omics – Towards Personalized Med for Rare Disorders. LPI; Amount: \$656k
5. Qatar-Morocco QNRF-CNRST Equine Research Grant (**QMRC01-0216-210004**): Building a resource for Equine Genome Research in Qatar and Morocco. LPI; Amount: \$400k
6. Sidra Internal Research Fund Cycle 4: ASPIRE COHORT OF NEWBORNS – Advancing newborn Screening Program to Identify babies at high Risk of future disorders for Effective Monitoring and Personalized Care. PI; Amount: \$275k
7. Breakthrough T1D (**4-SRA-2023-1375-M-B**): DIA-MENA: Type 1 Diabetes Islet Autoantibody Screening Program in the MENA: a Pilot-Cross Sectional prospective cohort study. PI; Amount: \$1M
8. National Priorities Research Program Cycle 13 (**NPRP13S-0121-200129**): Integrative omics, neuroimaging and cognition for early and specific biomarkers of ADHD. PI; \$580k
9. Sidra Internal Research Fund Cycle 3: The GENE-IUS Study – Genomic Investigation of Idiopathic and Undiagnosed Congenital Structural malformation in Qatar. LPI; Amount: 2m QAR
10. National Priorities Research Program Cycle 12 (**NPRP12-0318-190394**): From Molecular Etiology to Improved Reproductive Health for Male Infertility in Qatar (The FERTILITY-IQ Study). LPI; \$600k
11. National Priorities Research Program Cycle 11 (**NPRP11-0110-180250**): The PMED-Qatar Study – Personalized Molecular Evaluation and Diagnosis for Rare Diseases in Qatar. LPI; \$600k
12. Postdoctoral research award (**PDRA4-0218-18007**): National award for recruitment and mentorship of a Postdoctoral associate as part of our national Autism study. LPI; Grant amount: \$200K
13. Sidra Internal Research Fund Cycle 2: Establishment of Biobank, registry and genetic makeup of patients with cardiac defects and familial myocardiopathies. CoPI; Grant amount: 1m QAR
14. Sidra Internal Research Fund Cycle 2: Neurogenomics Model for Children affected with attention deficit Hyperactivity Disorder (ADHD) in the Qatari population. CoPI; Grant amount: 500K QAR
15. National Priorities Research Program Cycle 10 (**NPRP10-0202-170320**): The BARAKA-Qatar Study – Building A National Resource for the Advancement of Knowledge of Autism. LPI; \$1.4m
16. National Priorities Research Program Cycle 10 (**NPRP10-0202-170320**): Building haplotype-resolved *de novo* genome maps for Qatari family trios with severe developmental disorders as a premium resource to uncover etiology in the Qatari population. CoPI; Grant amount: \$1.25m
17. Sidra Internal Research Fund Cycle 1: The BRIDGE-WGS Program: Building a Robust End-to-End Clinical Research Disease Genetics with Whole Genome Sequencing. LPI; Amount: 1m QAR
18. Sidra Internal Research Fund Cycle 1: Epidemiology, Genetics and Management of T1D in Children in Qatar. CoPI; Grant amount: 1m QAR
19. Sidra Internal Fund Cycle 1: The PRECISION Study: Performing RNA-sequencing, Epigenetic and Cell-free nucleotide Investigation in Diabetic Retinopathy. CoPI; Amount: 500K QAR
20. Sidra Internal Research Fund Cycle 1: Landscape of Breast Cancer in Qatar; Amount: 500K QAR

21. Path towards Personalized Medicine Cycle 2 (**PPM2-0216-170014**): Genomics, Islamic Ethics and Public Engagement (GIEPE): Towards Bridging Knowledge Communication Gaps. CoPI; \$600k
22. Path towards Personalized Medicine Cycle 1 (**PPM-1-1913-3-396**): A high resolution map of structural variation in Qatari genomes & contribution to cardiovascular traits. LPI; \$966,334.01
23. National Priorities Research Program Cycle 9 (**NPRP-9-229-3-041**): The 1000 Qataromics Study: Validating molecular signature of diabetes retinopathy through genomic, proteomic and transcriptomic profiling in a thousand Qatari nationals. CoPI; Grant amount: \$971,002.18
24. National Research Priorities Program Cycle 8 (**NPRP-8-1913-3-396**): Design and implementation of the first Qatari population-centric high-throughput custom-genotyping microarray, with an accompanying private Qatari genome browser. CoPI; Grant amount: \$938,445.56

Publications and Scientific Seminars

Submitted and Accepted:

1. Biallelic variants in *CSPG4* cause a novel syndrome of intellectual disability, neurological impairments, and craniofacial anomalies. *Submitted*
2. The Qatar Mendelian Program – a national-scale precision medicine protocol for rare diseases. *Submitted*
3. Ensemble Multi-Trait Genomic Modelling Reveals Limitations of Polygenic Risk Scores and Improves Cross-Ancestry Prediction of Type 2 Diabetes. *In Review*
4. A catalogue of transcriptome genetic effects from a consanguineous Middle Eastern biobank. *In Review*
5. Examining the impact of mtDNA copy number and heteroplasmy on complex traits. *In Review*

Published:

1. Gandhi GD, Alenbawi J, Mbarek H, Fakhro KA, Nemer G. Population enrichment of ABCG5/ABCG8 variants in Qatar and genetic estimates of sitosterolemia burden. **J Clin Lipidol.** 2026 Apr 30:S1933-2874(26)00129-7. doi:10.1016/j.jacl.2026.04.029. Epub ahead of print. PMID: 42161750.
2. Ali H, Alahmad B, Marafi D, Alsharhan H, Al Sabah S, Fakhro KA, Lashuel HA, Alkuraya FS, Alsheikh-Ali A, Al-Mulla F, Abou Tayoun A. A genomics health strategy for the Arabian Gulf. **Nat Med.** 2026 May;32(5):1580-1582. doi: 10.1038/s41591-026-04276-y. PMID: 41781548.
3. Hashem S, Elhag SF, Awada Z, Aamer W, Bhat AA, Al-Maraghi A, Shaif A, Aldhaibani H, Fakhro K, Nemer G, Kamal M. Sex-Based Differences in Clinical and Social Outcomes in Qatari Youth with ADHD. **Child Psychiatry Hum Dev.** 2026 Feb 26. doi: 10.1007/s10578-026-01986-9. PMID: 41746460.
4. Aliyev E, Syed N, Visconti A, Aliyev T, Belkadi A, Ghorbani M, Rossi N, Naeem, H, Gandhi GD, Thareja G, Al-Maraghi A, Aamer W, Ibrahim AA, Shaath R, Al-Ajli, FO, Razali RM, Sedlazeck FJ, Davila S, Akil A, Suhre K, Mokrab Y, Falchi M, Fakhro KA. The biomedical landscape of genomic structural variation in the qatari population. **Nat Commun.** 2026 Jan 13;17(1):1019. doi: 10.1038/s41467-025-67763-9. PMID: 41530136; PMCID: PMC12847811.
5. Hashem S, Elhag SF, Bhat AA, Aamer W, Al-Maraghi A, Alhaboub H, Abuthaher D, Akil ASA, Haris M, Fakhro K, Nemer G, Kamal M. Functional Interpretation of a Novel Homozygous METTL5 Variant Associated with ADHD and Neurodevelopmental Abnormalities: A Case Report and Literature Review. **Genes (Basel).** 2025 Dec 15;16(12):1502. doi: 10.3390/genes16121502. PMID: 41465175.
6. Ibrahim AA, Aamer W, Aliyev E, Al-Maraghi A, Elhag SF, Abdi M, Syed N, Palaniswamy S, Akil ASA, Kamal M, Fakhro KA. Long-read sequencing identifies a novel de novo inversion in SMARCC2 in a pediatric patient with Coffin-siris syndrome 8: a case report. **BMC Med Genomics.** 2025 Nov 25;18(1):204. doi: 10.1186/s12920-025-02269-3. PMID: 41291750; PMCID: PMC12752245.

7. Ata M, Choucair F, Djekidel MN, Sousa Esteves M, Al Ali F, Nawaz S, El Taha L, Soloviov O, Sathappan A, Aliyev E, Fakhro K, Awwad J, Avella MA. Mutational disruption of transcription factors binding and regulatory networks in a case of unexplained total fertilization failure. **J Transl Med.** 2025 Nov 4;23(1):1224. doi: 10.1186/s12967-025-07038-5. PMID: 41188932; PMCID: PMC12584518.
8. Ahmed I, Bhat AA, Jeya SP, Wong WKM, Sadida HQ, Polkamp M, Hardikar HP, Lakshmi J, Hussain SA, Chin-Smith E, Ranjan AK, Joglekar MV, Hardikar AA, Fakhro K, Akil AA. Plasma multi-omics and machine learning reveal predictive biomarkers for type 2 diabetes and retinopathy in Qatar biobank cohort. **J Transl Med.** 2025 Oct 22;23(1):1159. doi: 10.1186/s12967-025-07113-x. PMID: 41126335.
9. Riera-Escamilla A, Arafa MM, Farnetani G, Xavier MJ, Oud MS, Majzoub AA, Ramos L, Abrardo C, Spinelli M, Defazio G, Ars E, Pybus M, Elbardisi HT, Nawaz S, Syed N, Ruiz-Castané E, van der Heijden GW, Fakhro KA, Veltman JA, Krausz C. Genetic determinants of testicular sperm extraction outcomes: insights from a large multicentre study of men with non-obstructive azoospermia. **Hum Reprod Open.** 2025 Aug 29;2025(3):hoaf049. doi:10.1093/hropen/hoaf049. PMID: 40896145
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 18. Devadoss Gandhi G, Aliyev E, Syed N, Vempalli FR, Saad C, Mbarek H, Al-Saei O, Al-Maraghi A, Abdi M, Badii R; Qatar Genome Program Research Consortium; Akil AA, Ben-Omran T, Fakhro KA. Mapping the genetic landscape of treatable inherited metabolic disorders in a large Middle Eastern biobank. **Genet Med.** 2024 Dec;26(12):101268. doi: 10.1016/j.gim.2024.101268. Epub 2024 Sep 14. PMID: 39286960.
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Contributed Book Chapters

1. Islamic Ethics and the Healthcare of Children in the Genetics Era – A WISH Report, 2022
2. A new Era for Vaccine Innovation – A WISH Report, 2022
3. Genes and Specific (related) proteins in Neurodevelopmental Disorders, 2022
4. Genomics of Autism, 2020
5. Genetic Evaluation of Male Infertility, 2019
6. Genomics in the Gulf Region and Islamic Ethics – A WISH Report, 2016

Invited Oral Presentations:

1. *Invited Panelist, The 3rd Congress of Medical Genetics*. Multi-omic approaches to improve diagnostic yield for rare diseases in Qatar. Madrid 2026

2. *Invited Panelist*, **BMJ Future Health – Middle East**. Beyond one-size fits all: the future of Precision Medicine. Doha 2026
3. *Invited Speaker*, **Alliance for Regenerative Medicine**. The landscape for Advanced Therapies in the Middle East. Rome 2026
4. *Invited Panelist*, **Cure Middle East Health Accelerator**. Building an innovation economy in the Gulf: the case for Qatar. New York 2025
5. *Invited Speaker*, **NASEM-KFAS 2nd Precision Medicine Workshop**. Navigating Policy and Regulation implementing Genomic Medicine in the GCC. Washington 2025
6. *Invited Speaker*, **1st Congress for Clinical & Translational Res.** Insights from Qatar. Abu Dhabi 2025
7. *Invited Speaker*, **Harvard Gulf Arab Association Conference**. AI in Healthcare, Boston 2025
8. *Invited Speaker*, **Izmir Biomedicine Genomics Symposium**. The Qatari Mendelian Program, Izmir 2025
9. *Invited Speaker*, **MENA Metabolic and Genetic Diseases Conference**. Integrating Metabolomics and Whole Genome Sequencing for Rare Disease Diagnostics and Treatment, Bahrain 2025
10. *Invited Speaker*, **Middle East Fertility Society**. The molecular basis of male infertility from whole genome sequencing, Qatar 2024
11. *Invited Speaker*, **United Nations General Assembly – Science Summit**. 2 panels: Genomics Research Ecosystem in Qatar; Advancing Genome Research for Underrepresented Populations, NY 2024
12. *Invited Speaker*, **Illumina Population Genetics Summit**. Qatar's Genomics Journey, UK 2024
13. *Invited Speaker*, **Regenerative Stem Cells Summit**. Rare disease genomics and precision medicine, Montenegro 2024
14. *Invited Speaker*, **Qatar University**. Governance and ethical considerations in genomics, Doha 2024
15. *Invited Speaker*, **Wellcome Trust Genomics of Rare Disease Conference**. Increasing diagnostic yield with multi-omics, UK 2024
16. *Invited Speaker*, **Medlab**. Advances in Genomics and Precision Medicine in the Qatar. Dubai 2024
17. *Invited Speaker*, **Liberty Fund**. Genomics and Public Health: Considerations of Liberty and Responsibility. London, UK 2023
18. *Invited Speaker*, **International Conference on Newborn Sequencing (ICoNS '23)**. The Qatari Newborn Sequencing Initiative. London, UK 2023
19. *Opening Plenary Keynote*, **Precision Medicine Expo & Summit**. Building Capacity for Genomic Medicine in the Arab World. Dubai, UAE 2023
20. *Invited Speaker*, **The 3rd Annual MENA Metabolic and Genetic Conference**. Genomic Medicine in 2023 – Challenges and Opportunities Ahead. Muscat, Oman 2023
21. *Invited Speaker*, **The 1st International UAE Rare Disease Society Congress**. Leveraging Multi-omic Approaches to Improve the Diagnostic Yield for Rare Diseases. Dubai, UAE 2023
22. *Invited Speaker*, **Doha-Heidelberg 3rd Rare Disease Conference**. Applications of Genomic Technologies in Rare Diseases – The Qatari Mendelian Experience. Doha, Qatar 2023
23. *Invited Showcase*, **Precision Medicine World Conference**. Building Future Precision Care in Qatar. San Jose, CA 2023
24. *Invited Showcase*, **Precision Medicine World Conference**. Population-Scale Genome Programs Fireside Chat (hosted by Microsoft). San Jose, CA 2023
25. *Invited Speaker*, **Sidra Pediatric Neuroscience Conference**. Towards Personalized Medicine in Autism Spectrum Disorder. Doha, Qatar 2022
26. *Closing Keynote Speaker*, **The 2nd Annual MENA Metabolic and Genetic Conference**. Debate: Whole genome sequencing to screen for genetic disorders across the age spectrum. Qatar (virtual) 2022

27. *Keynote Speaker, Precision Medicine Expo & Summit.* The Qatari Mendelian Program – Lessons learned solving rare diseases in consanguineous populations. Dubai, UAE 2022
28. *Opening Keynote Speaker, Pan Arab Human Genetics Conference.* Towards Precision Medicine at Scale for Arab Populations. Dubai, UAE (virtual) 2021
29. *Invited Speaker, Nations Genomes Congress 2021.* Genomics at a National Scale – The Qatari Experience. Russia (virtual) 2021
30. *Invited Speaker, Computational Biology and AI Symposium.* Algorithms and Approaches for the Diagnosis of Rare Disorders from NGS Datasets. Russia (virtual) 2021
31. *Invited Speaker, Weill-Cornell Medicine in Qatar Grand Rounds.* Population Genetics and Genomic Medicine in Qatar. Doha, Qatar 2020
32. *Invited Speaker, Nanyang Technological University.* Genomic Medicine in Qatar: Scientific, Social and Ethical considerations. Singapore 2020
33. *Invited Speaker, Weill-Cornell Medicine.* Population Disease Genetics in Qatar. NY, USA 2019
34. *Invited Speaker, Qatar University Biomedical Research Symposium.* Approaching the Inflection Point in the Precision Medicine Economy in Qatar. Doha, Qatar 2018
35. *Invited Speaker, World Innovation Summit in Health (WISH).* Building Capacity for Precision Medicine in Qatar. Doha, Qatar 2018
36. *Invited Speaker, Precision Medicine Beyond Cancer.* Towards Precision Medicine in Qatar – Opportunities for Academia and Industry Partnerships. Munich, Germany 2018
37. *Invited Speaker, Bilkent University Department of Molecular Genetics Seminar.* Sequencing of Mendelian Disorders in Qatar –the Path to Personalized Medicine. Ankara, Turkey 2018
38. *Invited Speaker, RIKEN Institute Seminar Series.* Delivering Next-Generation diagnostics to understudied global populations: the Qatari experience. Yokohama, Japan 2018
39. *Invited Speaker, Bellvitge Biomedical Research Institute (IDIBELL).* Mendelian and Complex Trait Analysis in the Qatari population. Barcelona, Spain, 2017
40. *Invited Speaker, Medical Genetics and Genomics in Health Care.* Building Qatari reference genomes to facilitate precision medicine in the Middle East. Kuwait City, Kuwait, 2016
41. *Invited Speaker, Genomics and Precision Medicine.* Extending the clinical utility of NGS beyond Mendelian genetics: the Qatar experience. Riyadh, Saudi Arabia, 2016
42. *Invited Speaker, Life of Genomes.* Adventures in Human genetics in Qatar. Kazan, Russia, 2016
43. *Invited Speaker, Genome Russia.* Large-scale Genomics in Qatar: Exploring Mendelian Disease and Population Genomics through Next Generation Sequencing. Moscow, Russia, 2016
44. *Accepted Oral Abstract, The 13th International Congress of Human Genetics.* A Population-Centric Reference Genome to Enable Precision Medicine in the Middle East. Kyoto, Japan 2016
45. *Invited Speaker, Sidra Functional Genomics Symposium.* Towards Personalized Medicine in Qatar through Mendelian and Population Genetics. Doha, Qatar, 2015
46. *Invited Speaker, Hamad Medical Corporation Grand Rounds.* Revolutionizing disease genetics in Qatar with next-generation sequencing. Doha, Qatar, 2014
47. *Invited Speaker, University of Southern California Pediatric Rounds.* Paving the ground for Personalized Medicine in Qatar through Population and Mendelian Studies. Los Angeles, CA, 2014
48. *Invited Speaker, Qatar Foundation Annual Research Conference.* Loss of Function Mutations in the Qatari Genome – Implications for Health and Disease. Doha, Qatar, 2013
49. *Accepted Oral Abstract, Human Genome Organization (HUGO).* Rare Structural Variants in Patients with Severe Congenital Heart Disease. Human Genome Meeting (Session: Advances in Genetics of Heritable Disorders). Dubai, United Arab Emirates, 2011

50. *Invited Speaker, Gordon Conference on Cilia and Mucociliary Interactions.* Copy Number Variations in Heterotaxy Patients Affect Genes Involved in Left-Right Patterning (Session: Genetic Approaches in Ciliopathies and Mucociliary Diseases). Ventura, CA, 2011